Question	All the energy required for life' processes is obtained by:
Type	multiple choice
Option	Reduction of some macromolecules
Option	Oxidation of all macromolecules
Option	Oxidation of some macromolecules
Option	Reduction of all macromolecules
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Choose the odd one out w.r.t respiratory substrates
Type	multiple choice
Option	Sucrose
Option	Proteins
Option	Vitamins
Option	Organic acids
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	All the energy contained in respiratory substrates is-
Type	multiple choice
Option	Released in a single step
Option	Used directly
Option	Released free into cell
Option	Not released free into the cell
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	The breakdown of complex molecules to yield energy takes place in -
Type	multiple choice
Option	Cytoplasm of eukaryotes
Option	Cytoplasm of prokaryotes
Option	Mitochondria of eukaryotes
Option	All of these
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	All living organisms retain-
Type	Enzymatic machinery to partially oxidise glucose without the help of oxygen
Option	Enzymatic machinery to partially oxidise glucose with the help of oxygen
Option	Enzymatic machinery to completely oxidise glucose with the help of oxygen
Option	Enzymatic machinery to completely oxidise glucose without the help of oxygen
Option	1
Answer	
Solution	
Positive Marks	4
Negative Marks	1
Question	S-I: Plants have specialized organs for gaseous exchange. S-II: Plants do not present great demands for gas exchange
Type	multiple choice

Option	Only S-I is correct
Option	Only S-II is correct
Option	Both S-I and S-II are correct
Option	Both S-I and S-II are incorrect
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Loose packing of parenchyma cells which provide an interconnected network of air spaces are present in-
Type	multiple choice
Option	Only leaves
Option	Only Stem and roots
Option	Leaves, stems and roots
Option	Only roots
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	The compounds that are oxidised during respiration process are known as
Type	multiple choice
Option	Respiratory quotient
Option	Respiratory product
Option	Oxidising product
Option	Respiratory substrate
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	How do the cells in the interior layers of a woody stem receive oxygen?
Type	multiple choice
Option	By diffusion
Option	Interior layers do not require oxygen as cells in interior tissues are dead
Option	All cells are living in the woody stem
Option	Interior layers get oxygen through lenticels
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	S-I: The process of breathing is very much connected to the process of release of energy. from food. S-II: Respiratory quotient depends upon the type of respiratory substrate used during respiration.
Type	multiple choice
Option	Only S-I is correct
Option	Only S-II is correct
Option	Both S-I and S-II are correct
Option	Both S-I and S-II are incorrect
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	In which of the following respiratory substrate, less CO_2 is evolved than the

	volume of O_2 consumed?
Type	multiple choice
Option	Sucrose
Option	Glucose
Option	Organic acid
Option	Fat
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	When proteins are respiratory substrates, the ratio would be about-
Type	multiple choice
Option	0.9
Option	1.0
Option	0.2
Option	0.7
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	RQ value of tripalmitin is-
Type	multiple choice
Option	1
Option	0.7
Option	1.2
Option	0.9
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following has R.Q. value more than one?
Type	multiple choice
Option	Carbohydrate
Option	Fat
Option	Protein
Option	Organic acids
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	When glucose is aerobically oxidized the RQ value will be-
Type	multiple choice
Option	One
Option	Zero
Option	More than one
Option	Less than one
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	The process which partially oxidise glucose without the help of oxygen is-
Type	multiple choice
Option	Krebs cycle

Option	Glycolysis
Option	Link reaction
Option	ETS
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	The similarity between Nostoc and Human cell is-
Type	multiple choice
Option	Naked nucleus present
Option	Krebs cycle not occur
Option	Glycolysis occurs in cytoplasm
Option	Glycolysis occurs in mitochondria
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	The number of ATP produced in Glycolysis is. ……
Type	multiple choice
Option	Three
Option	Four
Option	Two
Option	Six
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Number of ATP produced in aerobic Glycolysis is
Type	multiple choice
Option	Six
Option	Two
Option	Four
Option	Eight
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	S-I: In Glycolysis, ATP is utilized during conversion of phospho-enol pyruvate into pyruvate. S-II: Both Glucose and fructose can enter into glycolytic pathway.
Type	multiple choice
Option	Only S-I is correct
Option	Only S-II is correct
Option	Both S-I and S-II are correct
Option	Both S-I and S-II are incorrect
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	In glycolysis, NADH synthesis occurs at which step-
Type	multiple choice
Option	Conversion of PGAL into 1, 3 Diphosphoglycerate
Option	Conversion of 2- phosphoglycerate into phosphoenolpyruvate

Option	Conversion of glucose into glucose 6- phosphate
Option	Conversion of Diphosphoglycerate 1, 3 into 3- phosphoglycerate
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	In Glycolysis, ATP is produced in which steps -
Type	multiple choice
Option	Second and fifth
Option	Tenth only
Option	Fifth and seventh
Option	Seventh and tenth
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Mark the incorrect statement
Type	multiple choice
Option	Pyruvate kinase is involved in ATP synthesis in Glycolysis
Option	The term glycolysis has originated from the Latin words
Option	The scheme of glycolysis was given by Gustav Embden, Otto Meyerhof, and J. Parnas
Option	Sucrose is converted into fructose and glucose by enzyme invertase
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Isomerase enzyme converts -
Type	multiple choice
Option	PGAL to DiHAP
Option	Glucose - 6- phosphate to Fructose6-phosphate
Option	Fructose-6-phosphate to Fructose 1,6- bisphosphate
Option	Both (a) and (b)
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	The last enzyme of glycolysis which is involved in production of ATP is-
Type	multiple choice
Option	Phosphoglycerate kinase
Option	Pyruvate kinase
Option	Hexokinase
Option	All of these
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	The irreversible steps in Glycolysis is-
Type	multiple choice
Option	First only
Option	First and third
Option	Last
Option	Both (b) and (c)

Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Mark the incorrectly matched for (a) Only S-I is correct number of carbons
Type	multiple choice
Option	Phospho-enol pyruvate - 3C
Option	DIHAP - 6C
Option	Phosphoglycerate - 3C
Option	Glucose – 6- phosphate - 6C
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	S-I: In Fourth step of glycolysis, one 6- C compound break into two 3- C compound. S-II: The conversion of BPGA into 3- PGA is an energy yielding process.
Type	multiple choice
Option	Only S-I is correct
Option	Only S-II is correct
Option	Both S-I and S-II are correct
Option	Both S-I and S-II are incorrect
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	From the given examples- 1,3- BPGA, DiHAP, Pyruvate, Fructose, PGAL, PGA, PEP How many are sugars?
Type	multiple choice
Option	One
Option	Three
Option	Two
Option	Four
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	The products obtained from oxidation of one glucose in glycolysis are
Type	multiple choice
Option	Only Pyruvate
Option	One NADH + H^+, two ATP and one Pyruvate
Option	Two NADH + H^+, four ATP and one Pyruvate
Option	2 NADH + H^+, two ATP and two Pyruvate
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Complete oxidation of pyruvate in mitochondria involves process
Type	multiple choice
Option	Link reaction
Option	Glycolysis
Option	Krebs cycle

Option	Both (a) and (c)
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following occurs in link reaction?
Type	multiple choice
Option	Decarboxylation
Option	2-C compound is formed
Option	Oxidation
Option	All of these
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	All are the similarities between link reaction and krebs cycle, except
Type	multiple choice
Option	Location is mitochondrial matrix
Option	$FADH_2$ form
Option	Oxidative decarboxylation occurs
Option	NADH form
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Mark the incorrect matched pair w.r.t. number of carbons-
Type	multiple choice
Option	Acetyl CoA - 2C
Option	Oxalosuccinate - 6C
Option	Oxaloacetic acid -6C
Option	Succinate - 4C
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Pyruvate undergoes oxidative decarboxylation by a complex set of reactions catalyzed by enzyme
Type	multiple choice
Option	Pyruvate decarboxylase
Option	Pyruvate dehydrogenase
Option	Pyruvate kinase
Option	Isomerase
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	S-I: The TCA cycle starts with the condensation of acetyl group with oxalosuccinate and water to yield citric acid. S-II: The first product of TCA cycle is isomerized to form isocitrate
Type	multiple choice
Option	Only S-I is correct
Option	Only S-II is correct
Option	Both S-I and S-II are correct

Option	Both S-I and S-II are incorrect
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	In aerobic respiration, step in which complete oxidation of pyruvate by the stepwise removal of all the hydrogen atoms, leaving three molecules of CO_2 takes place in-
Type	multiple choice
Option	Outer membrane of mitochondria
Option	Matrix of mitochondria
Option	Inner membrane of mitochondria
Option	Both (b) and (c)
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	In one turn of Krebs cycle, number of NADH and $FADH_2$ formed are respectively-
Type	multiple choice
Option	Three and one
Option	Three and two
Option	Two and one
Option	One and Two
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	The correct statement related to the first product of Krebs cycle is-
Type	multiple choice
Option	It has three carboxylic acid group
Option	It is 6-C compound
Option	It is formed as a result of condensation process
Option	All of these
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following compound undergoes two successive steps of decarboxylation to give Succinyl coA?
Type	multiple choice
Option	Isocitrate
Option	Citrate
Option	$\propto -$ ketoglutarate
Option	OAA
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Oxalosuccinate is converted into aketoglutaric acid by-
Type	multiple choice
Option	Oxidation
Option	Oxidative decarboxylation

Option	Decarboxylation
Option	Reduction
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	S-I: $\propto -$ketoglutarate is converted into Succinyl coA by oxidative decarboxylation S-II: Substrate level phosphorylation occurs during the conversion of succinyl-CoA to succinic acid.
Type	multiple choice
Option	Only S-I is correct
Option	Only S-II is correct
Option	Both S-I and S-II are correct
Option	Both S-I and S-II are incorrect
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Coenzyme required by enzyme pyruvate dehydrogenase in link reaction is-
Type	multiple choice
Option	Thiamine pyrophosphate (TPP)
Option	NAD^+
Option	Coenzyme A
Option	All of these
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	How many steps in the Krebs cycle involves decarboxylation?
Type	multiple choice
Option	Two
Option	One
Option	Three
Option	Six
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	In aerobic respiration, step in which the passing on of the electrons removed as part of the hydrogen atoms to molecular O_2 with simultaneous synthesis of ATP takes place in-
Type	multiple choice
Option	Outer membrane of mitochondria
Option	Matrix of mitochondria
Option	Inner membrane of mitochondria
Option	Both (b) and (c)
Answer	3
Solution	
Positive Marks	4
Negative Marks	1